

Question 1

Consider the following optimization problem,

$$\begin{array}{ll}\text{minimize} & x^2 + 1 \\ \text{subject to} & (x - 2)(x - 4) \leq 0\end{array}$$

with variable $x \in \mathbf{R}$.

1. *Analysis of primal problem.* Give the feasible set, the optimal value, and the optimal solution.

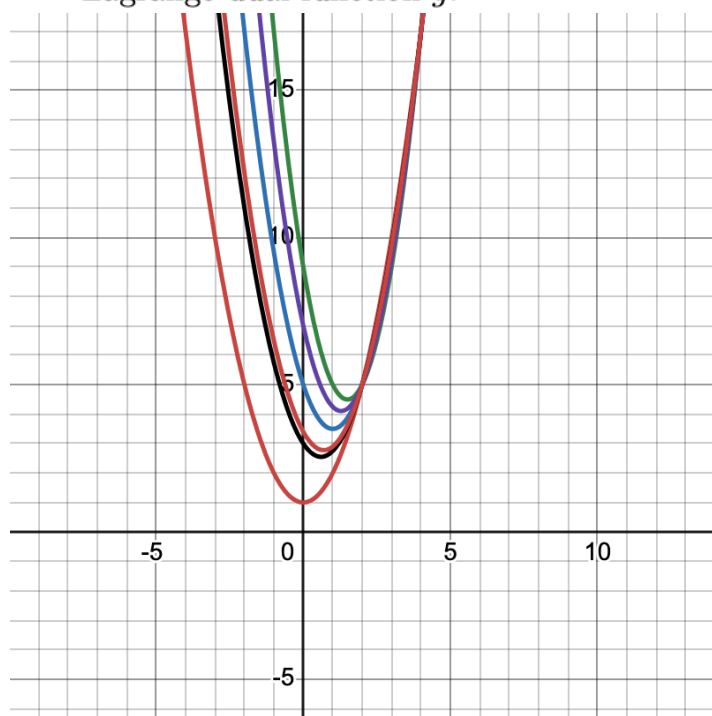
the feasible set is $[2, 4]$

the objective $x^2 + 1$ is increasing on $[2, 4]$.

$\Rightarrow$ minimum point is 2, optimal value $p = 2^2 + 1 = 5$

$\Rightarrow \begin{cases} \text{feasible set } [2, 4] \\ \text{optimal solution: } x^* = 2 \\ \text{optimal value: } p^* = 5 \end{cases}$

2. *Lagrangian and dual function.* Plot the objective $x^2 + 1$ versus x . On the same plot, show the feasible set, optimal point and value, and plot the Lagrangian $L(x, \lambda)$ versus x for a few positive values of λ . Verify the lower bound property ($p^* \geq \inf_x L(x, \lambda)$ for $\lambda \geq 0$). Derive and sketch the Lagrange dual function g .



Lagrangian plots

$\lambda = 0.25$: black

$\lambda = 0.3$: red

$\lambda = 0.5$: blue

$\lambda = 0.75$: purple

$\lambda = 1$: green

$$L(x, \lambda) = x^2 + 1 + \lambda(x - 2)(x - 4) = (1 + \lambda)x^2 - 6\lambda x + (8\lambda + 1) \quad \lambda \geq 0$$

$$g(\lambda) = \inf_x L(x, \lambda) \quad \text{since } 1 + \lambda > 0, \text{ the minimizer of } L(x, \lambda) \text{ over } x$$
$$\text{is } x(\lambda) = \frac{3\lambda}{1 + \lambda}$$

$$\Rightarrow g(\lambda) = \frac{1+9\lambda-\lambda^2}{1+\lambda} \leq p^* \Rightarrow \text{this verifies the lower bound}$$

3. *Lagrange dual problem.* State the dual problem, and verify that it is a concave maximization problem. Find the dual optimal value and dual optimal solution λ^* . Does strong duality hold?

$$\text{dual problem: maximize } g(\lambda) = \frac{1+9\lambda-\lambda^2}{1+\lambda}, \lambda \geq 0.$$

because g is a dual function, it's concave

to find the maximizer, differentiate:

$$g'(\lambda) = \frac{8-2\lambda-\lambda^2}{(1+\lambda)^2}$$

$$g'(\lambda) = 0 \Rightarrow \lambda^* = 2 \text{ (feasible)}$$

$$\Rightarrow d^* = g(2) = 5.$$

$\Rightarrow$ dual optimal solution $\lambda^* = 2$, dual optimal value $d^* = 5$

$\Rightarrow d^* = p^* = 5 \Rightarrow$ strong duality holds.

4. *Sensitivity analysis.* Let $p^*(u)$ denote the optimal value of the problem

$$\begin{array}{ll} \text{minimize} & x^2 + 1 \\ \text{subject to} & (x-2)(x-4) \leq u \end{array}$$

as a function of parameter u . Plot $p^*(u)$. Verify that $dp^*(0)/du = -\lambda^*$.

$$\text{let } p^*(u) = \min_x \{x^2 + 1 \mid (x-2)(x-4) \leq u\}$$

$$\Rightarrow x = 3 \pm \sqrt{1+u}.$$

$\Rightarrow$ when $u \geq -1$, feasible set $[3-\sqrt{1+u}, 3+\sqrt{1+u}]$

$\Rightarrow$ the minimizer is the feasible point with minimal abs.

$$\text{for } -1 \leq u \leq 8, \quad x^*(u) = 3 - \sqrt{1+u} \Rightarrow p^*(u) = 10+u - 6\sqrt{1+u}$$

$$\text{for } u \geq 8 \quad \cdot \quad x^* = 0 \Rightarrow p^*(u) = 1$$

$$\text{for } u < -1, \quad \text{infeasible.} \quad p^*(u) = +\infty$$

$$\Rightarrow p^*(u) = \begin{cases} +\infty, & u < -1 \\ 10+u - 6\sqrt{1+u}, & -1 \leq u \leq 8 \\ 1, & u \geq 8 \end{cases}$$

$$\Rightarrow \left. \frac{d}{du} p^*(u) \right|_{u=0} = -2. \quad \text{since } x^* = 2,$$

$$\Rightarrow \left. \frac{d}{du} p^*(u) \right|_{u=0} = -\lambda^*$$

Question 2

Express the dual problem of

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & f(x) \leq 0, \end{array}$$

with $c \neq 0$, in terms of the conjugate f^* . Explain why the problem you give is convex. We do not assume f is convex.

$$L(x, \lambda) = c^T x + \lambda f(x), \quad \lambda \geq 0. \quad g(\lambda) = \max_x (c^T x + \lambda f(x))$$

$$\text{for } \lambda > 0, \quad g(\lambda) = -\lambda \sup_x \left[\left(-\frac{c}{\lambda}\right)^T x - f(x) \right]$$

$$\Rightarrow g(\lambda) = -\lambda f^*\left(-\frac{c}{\lambda}\right), \quad \lambda > 0$$

$$\text{if } \lambda = 0, \quad g(0) = -\infty$$

$$\Rightarrow \text{the dual problem is maximize } -\lambda f^*\left(-\frac{c}{\lambda}\right) \text{ s.t. } \lambda > 0.$$

this is convex opt. problem because it's a max of a concave function over a convex set. f^* is always convex, $\lambda f^*\left(-\frac{c}{\lambda}\right)$ is a perspective of a convex function. $\Rightarrow$ it's negative is concave.

Question 3

Consider the following LP.

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

Give the dual problem, and make the implicit equality constraints explicit.

$$\begin{aligned}L(x, \lambda, v) &= c^T x + \lambda^T (Gx - h) + v^T (Ax - b), \lambda \geq 0 \\ &= (c + G^T \lambda + A^T v)^T x - h^T \lambda - b^T v\end{aligned}$$

$\Rightarrow$ the dual is finite iff $c + G^T \lambda + A^T v = 0$

$$\Rightarrow g(\lambda, v) = -h^T \lambda - b^T v$$

$\Rightarrow$ the dual problem is maximize $-h^T \lambda - b^T v$.
s.t. $G^T \lambda + A^T v + c = 0, \lambda \geq 0$

$$\begin{aligned}\Rightarrow \text{maximize} & -h^T \lambda - b^T v^+ + b^T v^- \\ \text{s.t.} & G^T \lambda + A^T v^+ - A^T v^- + c = 0 \\ & \lambda \geq 0, v^+ \geq 0, v^- \geq 0\end{aligned}$$

Question 4

Consider the convex piecewise-linear minimization problem,

$$\text{minimize } \max_{i=1, \dots, m} (a_i^T x + b_i) \quad (1)$$

with variable $x \in \mathbf{R}^n$.

1. Derive a dual form based on the Lagrange dual of the equivalent problem

$$\begin{array}{ll} \text{minimize} & \max_{i=1, \dots, m} y_i \\ \text{subject to} & a_i^T x + b_i = y_i, \quad i = 1, \dots, m \end{array}$$

with variable $x \in \mathbf{R}^n$ and $y \in \mathbf{R}^m$.

$$\begin{aligned} L(x, y, \lambda) &= \max_i y_i + \sum_{i=1}^m \lambda_i (a_i^T x + b_i - y_i) \\ &= \max_i y_i - \lambda^T y + \lambda^T b + x^T \left(\sum_{i=1}^m \lambda_i a_i \right) \end{aligned}$$

$$\inf_{x, y} L \text{ finite} \Rightarrow \sum_{i=1}^m \lambda_i a_i = 0,$$

$$\inf_y (\max_i y_i - \lambda^T y) > -\infty \Rightarrow \lambda_i \geq 0, \quad \sum_{i=1}^m \lambda_i = 1$$

$$\Rightarrow \text{dual is } g(\lambda) = \begin{cases} b^T \lambda, & \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1, \sum_{i=1}^m \lambda_i a_i = 0 \\ -\infty, & \text{otherwise} \end{cases}$$

$$\begin{aligned} \Rightarrow \text{dual problem: } & \text{maximize } b^T \lambda \\ \text{s.t. } & \sum_{i=1}^m \lambda_i a_i = 0, \quad \sum_{i=1}^m \lambda_i = 1, \quad \lambda_i \geq 0 \end{aligned}$$

2. Formulate the piecewise-linear minimization problem (1) as an LP, and form the dual of the LP. Relate the LP dual to the dual obtained in part (1).

$$\text{minimize } t$$

$$\text{s.t. } a_i^T x + b_i \leq t.$$

$$\Rightarrow L(x, t, \lambda) = t + \sum_{i=1}^m \lambda_i (a_i^T x + b_i - t), \lambda_i \geq 0$$

$$= x^T \left(\sum_{i=1}^m \lambda_i a_i \right) + t \left(1 - \sum_{i=1}^m \lambda_i \right) + b^T \lambda.$$

$$\text{finite} \Rightarrow \sum_{i=1}^m \lambda_i a_i = 0, \quad \sum_{i=1}^m \lambda_i = 1$$

$$\Rightarrow g(\lambda) = b^T \lambda$$

$$\Rightarrow \text{LP dual: } \max. \quad b^T \lambda$$

$$\text{s.t. } \sum_{i=1}^m \lambda_i a_i = 0, \quad \sum_{i=1}^m \lambda_i = 1, \quad \lambda_i \geq 0$$

Question 5

Analytical centering. Derive a dual form for

$$\text{minimize} \quad - \sum_{i=1}^m \log(b_i - a_i^T x)$$

with domain $\{x \mid a_i^T x < b_i, i = 1, \dots, m\}$. First introduce new variables y_i and equality constraints $y_i = b_i - a_i^T x$.

$$\text{let } y_i = b_i - a_i^T x. \Rightarrow y_i > 0 \quad \& \quad a_i^T x + y_i = b_i$$

$$\Rightarrow \text{minimize} \quad - \sum_{i=1}^m \log y_i$$

$$\text{s.t. } a_i^T x + y_i = b_i, \quad y_i > 0$$

$$\Rightarrow L(x, y, v) = - \sum_{i=1}^m \log y_i + v^T (Ax + y - b) = x^T A^T v + \sum_{i=1}^m (-\log y_i + v_i y_i) - b^T v$$

let $\inf_x$ be finite, $\Rightarrow A^T v = 0$.

for each i , $\inf_{y_i > 0} (-\log y_i + v_i y_i)$ is finite iff. $v_i > 0$.

$$\Rightarrow -\frac{1}{y_i} + v_i = 0 \Rightarrow y_i = \frac{1}{v_i}$$

$$\Rightarrow g(v) = \sum_{i=1}^m (\log v_i + 1) - b^T v \quad (v > 0, A^T v = 0)$$

$$\Rightarrow \text{dual problem is } \max_{v > 0} m + \sum_{i=1}^m \log v_i - b^T v$$
$$\text{s.t. } A^T v = 0, v_i > 0$$

Question 6

Consider the the convex optimization problem,

$$\begin{array}{ll} \text{minimize} & e^{-x} \\ \text{subject to} & x^2/y \leq 0 \end{array}$$

with variable x and y , and domain $\mathcal{D} = \{(x, y) \mid y > 0\}$.

1. Verify that this problem is a convex problem and find the optimal solution and value.

e^{-x} convex, $\frac{x^2}{y}$ is convex on $y > 0$ (perspective)

$$\Rightarrow x=0 \Rightarrow \text{feasible set is } \{(0, y) \mid y > 0\}.$$

$$\Rightarrow \text{for every feasible point. } e^{-0} = 1$$

$$\Rightarrow p^* = 1 \Rightarrow \text{optimal solution is every point in the feasible set.}$$
$$\text{optimal value } p^* = 1$$

2. Give the Lagrange dual problem, and find the optimal solution λ^* and optimal value d^* of the dual problem. What is the optimal duality gap?

$$L(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y}, \quad \lambda \geq 0.$$

$$g(\lambda) = \max_{x, y > 0} (e^{-x} + \lambda \frac{x^2}{y})$$

for any x , if $\lambda > 0$, let $y \rightarrow +\infty$, $\Rightarrow \lambda \frac{x^2}{y} \rightarrow 0$

$$\Rightarrow g(\lambda) = 0.$$

if $\lambda = 0$, $g(\lambda) = 0$

$\Rightarrow$ dual problem: $\max. 0$
s.t. $\lambda \geq 0$

$$\Rightarrow d^* = 0$$

any $\lambda^* \geq 0$ is dual optimal. $\Rightarrow$ gap is $p^* - d^* = 1$.

3. Does Slater's condition hold for this problem?

Slater's condition requires $\frac{x^2}{y} < 0$.

$\Rightarrow$ not hold

4. What is the optimal value $p(u)$ of the perturbed problem

$$\begin{array}{ll} \text{minimize} & e^{-x} \\ \text{subject to} & x^2/y \leq u \end{array}$$

as a function of u ? Verify that the global sensitivity inequality

$$p^*(u) \geq p^*(0) - \lambda^* u$$

does not hold.

$$\text{let } p^*(u) = \inf \{ e^{-x} \mid \frac{x^2}{y} \leq u, y > 0 \}$$

$$\text{if } u < 0, \text{ infeasible, } \Rightarrow p^*(u) = +\infty$$

$$\text{if } u = 0, \quad x = 0, \Rightarrow p^*(u) = 1$$

$$\text{if } u > 0 \quad \text{for any } x, \quad y \geq \frac{x^2}{u}, \quad p^*(u) = \inf_x e^{-x} = 0.$$

$$\Rightarrow p^*(u) = \begin{cases} +\infty, & u < 0 \\ 1, & u = 0 \\ 0, & u > 0 \end{cases}$$

$$p^*(u) \geq p^*(0) - \lambda^* u. \quad \text{let } \lambda^* = 0, \Rightarrow p^*(u) \geq 1.$$

$$\text{while for } u > 0, \quad p^*(u) = 0 < 1$$

$\Rightarrow$ does not hold.

Question 7

Consider the QCQP,

$$\begin{array}{ll} \text{minimize} & x_1^2 + x_2^2 \\ \text{subject to} & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \quad \textcircled{1} \\ & (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \quad \textcircled{2} \end{array}$$

with variable $x \in \mathbf{R}^2$.

1. Sketch the feasible set and level sets of the objective. Find the optimal point x^* and optimal value p^* .

$$\begin{aligned} \textcircled{1} - \textcircled{2} &\Rightarrow (x_2 - 1)^2 - (x_2 + 1)^2 \leq 0 \\ \textcircled{2} - \textcircled{1} &\Rightarrow (x_2 + 1)^2 - (x_2 - 1)^2 \leq 0 \end{aligned} \quad \left| \Rightarrow x_2 = 0 \right.$$

$$\Rightarrow x_1 = 1, x_2 = 0. \Rightarrow \text{feasible set } \{(1, 0)\}.$$

$$\Rightarrow \text{optimal point } x^* = (1, 0), \text{ optimal value } p^* = 1$$

2. Give the KKT conditions. Do there exist Lagrange multipliers λ_1^* and λ_2^* that prove x^* is optimal?

$$\text{let } f_1(x) = \textcircled{1}, f_2(x) = \textcircled{2}$$

$$\text{KKT: 1) primary feasibility: } f_1(x^*) \leq 0, f_2(x^*) \leq 0$$

$$2) \text{ dual feasibility: } \lambda_1^* \geq 0, \lambda_2^* \geq 0$$

$$3) \text{ complementary slackness: } \lambda_1^* f_1(x^*) = 0, \lambda_2^* f_2(x^*) = 0$$

$$4) \text{ stationarity: } \nabla(x_1^2 + x_2^2) + \lambda_1^* \nabla f_1(x^*) + \lambda_2^* \nabla f_2(x^*) = 0$$

$$\nabla(x_1^2 + x_2^2) = (2x_1, 2x_2) \Rightarrow \text{at } x^* = (1, 0), \nabla = (2, 0)$$

$$\nabla f_1(x) = (2(x_1 - 1), 2(x_2 - 1)) \Rightarrow \nabla f_1 = (0, -2)$$

$$\text{similarly } \nabla f_2 = (0, 2)$$

$$\text{sta: } (2, 0) + \lambda_1^*(0, -2) + \lambda_2^*(0, 2) = 0 \Rightarrow 2 = 0, \text{ impossible}$$

$$\Rightarrow \text{none Lagrangian}$$

3. Derive and solve the Lagrange dual problem. Does strong duality hold?

$$L(x, \lambda_1, \lambda_2) = x_1^2 + x_2^2 + \lambda_1 f_1(x) + \lambda_2 f_2(x), \quad \lambda_1, \lambda_2 \geq 0$$

$$\text{let } s = \lambda_1 + \lambda_2. \quad L(x, \lambda_1, \lambda_2) = (4s)(x_1^2 + x_2^2) - 2sx_1 + 2(\lambda_2 - \lambda_1)x_2 + s$$

min over x_1, x_2

$$\Rightarrow x_1 = \frac{s}{1+s}, \quad x_2 = \frac{\lambda_1 - \lambda_2}{1+s}$$

$$\Rightarrow g(\lambda_1, \lambda_2) = \frac{\lambda_1 + \lambda_2 - (\lambda_1 - \lambda_2)^2}{\lambda_1 + \lambda_2 + 1} \quad (\lambda_1, \lambda_2 \geq 0)$$

$$\Rightarrow \text{dual problem: } \max_{\lambda_1, \lambda_2} \frac{\lambda_1 + \lambda_2 - (\lambda_1 - \lambda_2)^2}{1 + \lambda_1 + \lambda_2}$$

$$\text{s.t. } \lambda_1 \geq 0, \lambda_2 \geq 0$$

for λ_1, λ_2 . when $\lambda_1 = \lambda_2$, the numerator max

$$\text{let } \lambda_1 = \lambda_2 = t \geq 0, \text{ then } g(t, t) = \frac{2t}{1+2t}$$

$$\Rightarrow d^* = \sup_{\lambda_1, \lambda_2 \geq 0} g(\lambda_1, \lambda_2) = 1$$

since $p^* = d^* = 1 \Rightarrow$ hold in value

the dual optimum not attained at any finite (λ_1, λ_2)

it's near $\lambda_1 = \lambda_2 \rightarrow \infty$.